MUHID HASSAN RISVY

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GitHub ⋄ *LinkedIn*

EDUCATION

New Jersey Institute of Technology — Fall 2024 - Current
Ph.D. (Ongoing) in Computer Science

Shahjalal University of Science & Technology — May 2023
B.Sc. in Computer Science and Engineering
CGPA: 3.56/4.00 (**3.85** in the last two years).

RESEARCH INTERESTS AND EXPERIENCES

Research Interests: Cybersecurity, Human-Centric Security and Privacy, Human Computer Interaction.

Security Analysis of Passkey — Fall 2025-
Currently analyzing how passkeys are saved in users' password managers. Analyzed credential storage security across multiple password managers and found significant metadata leakage, weak encryption technique which can reveal user's information like login data, URL visited, timestamps and other critical data. Moreover, developed a browser extension that shows compromised or at-risk passkeys for each relying party, i.e, website.

Browser Automation For Improving Security Guides — Summer 2025
Developed an AI-driven system to automatically validate and update security guides from organizations like EFF when platforms change their UI or settings. Built a hybrid local/remote browser automation system using open-source LLM models deployed on HPC clusters. Successfully integrated remote HPC GPU resources with local browser operations and used Playwright for secure web automation.

Detecting Typosquatting Attack on the Web — Fall 2024 - Spring 2025
The research domain focused on establishing the authenticity of software on the internet, with emphasis on typosquatting, a cybercrime where attackers register domains that are typo-versions of legitimate domains. I explored and analyzed defensive strategies and mitigation challenges, as well as the gaps in existing research.

Security Analysis of Software-Based Cryptocurrency Wallets: (Undergraduate Thesis): The research project aims to identify the security issues and privacy concerns in cryptocurrency wallets. It uses vulnerability assessment tools and user reviews to evaluate the perceived security and usability. Both manual and machine learning approaches are used to analyze user reviews and determine the wallets' effectiveness in keeping cryptocurrencies secure.

TEACHING EXPERIENCE

Teaching Assistant — Fall 2024 - Current
New Jersey Institute of Technology

CS 351	Introduction to Cybersecurity	Fall 2024
CS 388	Android Application Development	Spring 2025
CS 288	Intensive Programming in Linux	Summer 2025
CS 332	Principles of Operating Systems	Summer 2025
CS 288	Intensive Programming in Linux (two sections)	Fall 2025

PROJECTS

COVID-19 Vaccine Supply Chain | *GitHub*
A blockchain project implementing a private blockchain-powered supply chain to establish end-to-end traceability for the COVID-19 vaccine, while guaranteeing secure distribution and transportation.
Technology: Hyperledger Fabric, Node.js, JavaScript, Apache CouchDB.

Passkey Health Manager | *GitHub*
A Chrome extension that manages passkeys with advanced health auditing features to track passkey security, detect weak keys using hash collision detection, monitor share counts, and maintain comprehensive export history.
Technology: React, TypeScript, Chrome Extension API.

E-commerce & Bank | *GitHub*
API-based website to simulate the functionalities of e-commerce services among 3 different organizations i.e., Customer, Supplier, and a Bank for instant transactions.
Technology: Node.js, Express.js, MongoDB.

RELATED COURSES AND SKILLS

Courses	CS 698: Human Factors in Security and Privacy (A), CS 756: Mobile Computing and Sensor Networks (A), CS 645: Security and Privacy in Computer Systems (A)
Programming	Solved 850+ problems on *CodeForces*, *LeetCode*, *Vjudge*, etc.
Languages	C, C++, Java, Python, JavaScript, MATLAB

ACCOMPLISHMENTS

Bangladesh Blockchain Olympiad 2022 June 2022
Finalist

Bangladesh Blockchain Olympiad 2021 February 2021
Finalist

EXTRA-CURRICULAR ACTIVITIES

Soccer: Champion, Intra IICT Tournament, 2019

Table Tennis: Runners-Up, Intra-Department Tournament, 2022